# Revision Game Cards — 1.1 Processors, Input, Output & Storage

Print and cut along the borders. ✂️ Loop cards: each player reads the bottom "Who has…?"; whoever holds the matching "I have…" reads it out, then their own clue — the chain visits every card and loops back to the start.

## Loop cards (dominoes) — cut out & deal

1 / 16

I HAVE

**Register**

WHO HAS…

the small, fast memory on or near the CPU that holds frequently used data to cut slow RAM fetches?

2 / 16

I HAVE

**Cache**

WHO HAS…

the technique of overlapping the fetch, decode and execute of consecutive instructions to raise throughput?

3 / 16

I HAVE

**Pipelining**

WHO HAS…

an independent processing unit able to run its own fetch-decode-execute cycle?

4 / 16

I HAVE

**Core**

WHO HAS…

the many-core processor that performs the same operation on large data sets in parallel?

5 / 16

I HAVE

**GPU**

WHO HAS…

carrying out multiple computations simultaneously by splitting a task across cores?

6 / 16

I HAVE

**Parallel processing**

WHO HAS…

the architecture with one shared memory and bus for both data and instructions?

7 / 16

I HAVE

## Von Neumann

WHO HAS...

the architecture with separate memories and buses for data and instructions, allowing simultaneous access?

8 / 16

I HAVE

## Harvard

WHO HAS...

the processor type with many complex, variable-length instructions and complexity in hardware (e.g. x86)?

9 / 16

I HAVE

## CISC

WHO HAS...

the processor type with few simple fixed-length instructions aiming for one per cycle, used in ARM mobile devices?

10 / 16

I HAVE

## RISC

WHO HAS...

the CPU component that performs arithmetic, logic and shift operations?

11 / 16

I HAVE

## ALU

WHO HAS...

the CPU component that decodes instructions, generates control signals and synchronises with the clock?

12 / 16

I HAVE

## Control Unit

WHO HAS...

the register holding the address of the next instruction, incremented each cycle?

13 / 16

I HAVE

## Program Counter (PC)

WHO HAS...

the register that stores the immediate result of an ALU calculation?

14 / 16

I HAVE

## Accumulator (ACC)

WHO HAS...

the use of secondary storage as extra RAM when physical RAM is full?

15 / 16

I HAVE

**Virtual memory**

WHO HAS...

the property of memory that loses its contents when power is removed (e.g. RAM)?

16 / 16

I HAVE

**Volatile**

WHO HAS...

the small, very fast store inside the CPU that holds data during processing?

## Taboo cards — describe the term without the banned words

### MAR (Memory Address Register)

Don't say:
- address
- location
- memory
- read

### Cache

Don't say:
- fast
- CPU
- frequently
- RAM

### Pipelining

Don't say:
- overlap
- fetch
- execute
- instructions

### Von Neumann

Don't say:
- shared
- memory
- bus
- instructions

### GPU

Don't say:
- parallel
- cores
- graphics
- pixels

### Virtual memory

Don't say:
- RAM
- secondary
- disk
- pages

### Volatile

Don't say:
- power
- lost
- RAM
- contents

### Optical storage

Don't say:
- laser
- CD
- DVD
- pits

## Bingo — word bank & ready-to-cut cards

**Word bank** (students fill a blank 3×3 grid, or use the pre-filled cards below): ALU, Control Unit, Program Counter, MAR, MDR, CIR, Accumulator, Cache, Pipelining, Core, GPU, Clock speed, Von Neumann, Harvard, CISC, RISC, Magnetic storage, Flash/SSD, Optical storage, Virtual memory

**BINGO 1**

| | | |
|---|---|---|
| Harvard | MAR | Von Neumann |
| Clock speed | Program Counter | Cache |
| RISC | Pipelining | Accumulator |

**BINGO 2**

| | | |
|---|---|---|
| Control Unit | Accumulator | Program Counter |
| Von Neumann | CIR | Optical storage |
| Clock speed | RISC | Core |

**BINGO 3**

| | | |
|---|---|---|
| Magnetic storage | Harvard | Cache |
| MDR | Virtual memory | RISC |
| Flash/SSD | Program Counter | GPU |

## BINGO 4

| | | |
|---|---|---|
| RISC | Flash/SSD | Clock speed |
| Cache | CISC | Magnetic storage |
| Pipelining | Virtual memory | Accumulator |

## BINGO 5

| | | |
|---|---|---|
| Harvard | MAR | Flash/SSD |
| Magnetic storage | Accumulator | Von Neumann |
| Virtual memory | CISC | Optical storage |

## BINGO 6

| | | |
|---|---|---|
| RISC | Von Neumann | Flash/SSD |
| Harvard | Core | CIR |
| Pipelining | GPU | Program Counter |

## Bingo — caller's list (teacher: read the clue, students cross off the term)

| # | Clue to read aloud | Answer |
|---|---|---|
| 1 | Performs arithmetic, logic and shift operations. | ALU |
| 2 | Decodes instructions and generates control signals. | Control Unit |
| 3 | Holds the address of the next instruction. | Program Counter |
| 4 | Holds the address of the location to read from or write to. | MAR |
| 5 | Holds the data or instruction just fetched from memory. | MDR |
| 6 | Holds the current instruction while it is decoded and executed. | CIR |
| 7 | Stores the immediate result of an ALU calculation. | Accumulator |
| 8 | Small fast memory on/near the CPU holding frequently used data. | Cache |
| 9 | Overlapping fetch, decode and execute to raise throughput. | Pipelining |
| 10 | An independent unit that runs its own FDE cycle. | Core |
| 11 | Many-core processor applying one operation across large data sets. | GPU |
| 12 | Pulses per second that synchronise the CPU, measured in Hz. | Clock speed |
| 13 | One shared memory and bus for data and instructions. | Von Neumann |
| 14 | Separate memories and buses for data and instructions. | Harvard |
| 15 | Many complex, variable-length instructions; complexity in hardware. | CISC |
| 16 | Few simple fixed-length instructions; used in ARM devices. | RISC |
| 17 | Stores data as magnetised patterns on a spinning platter. | Magnetic storage |
| 18 | Non-volatile EEPROM storage with no moving parts. | Flash/SSD |
| 19 | A laser reads pits and lands via reflected light. | Optical storage |
| 20 | Uses secondary storage as extra RAM when RAM is full. | Virtual memory |

## Quick-fire — clue & answer (self-test / quiz-quiz-trade)

| # | Clue | Answer |
|---|---|---|
| 1 | The bus that is unidirectional, carrying addresses from CPU to memory. | Address bus |
| 2 | The register pair you must never swap: address vs data. | MAR and MDR |
| 3 | The step in fetch most often forgotten, along with copying MDR to CIR. | Incrementing the PC |
| 4 | Non-volatile memory holding the bootstrap/BIOS firmware. | ROM |
| 5 | Volatile memory holding currently running programs and data. | RAM |
| 6 | The cache level that is fastest and smallest, on the core. | L1 cache |
| 7 | Excessive paging between RAM and disk that grinds a system to a halt. | Disk thrashing |
| 8 | The part of a task that must run sequentially, limiting the speed-up from adding cores. | Serial (sequential) fraction |
| 9 | Storage access type of tape: you must read through in sequence. | Serial (sequential) access |
| 10 | Technique flash memory uses to spread writes and prolong cell life. | Wear levelling |
| 11 | Output device that builds objects layer by layer. | 3D printer |
| 12 | The general rule for 'choose a device' questions: link each characteristic to the requirement. | Scenario / requirement (and compare) |